

Cost Analysis

2026-05-03

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1 Assumptions

The aim of the analysis is to identify the most cost-effective production periods in months of data and recommend optimal operating windows for the corresponding key influencers that can be set as a benchmark for the weekly analysis.

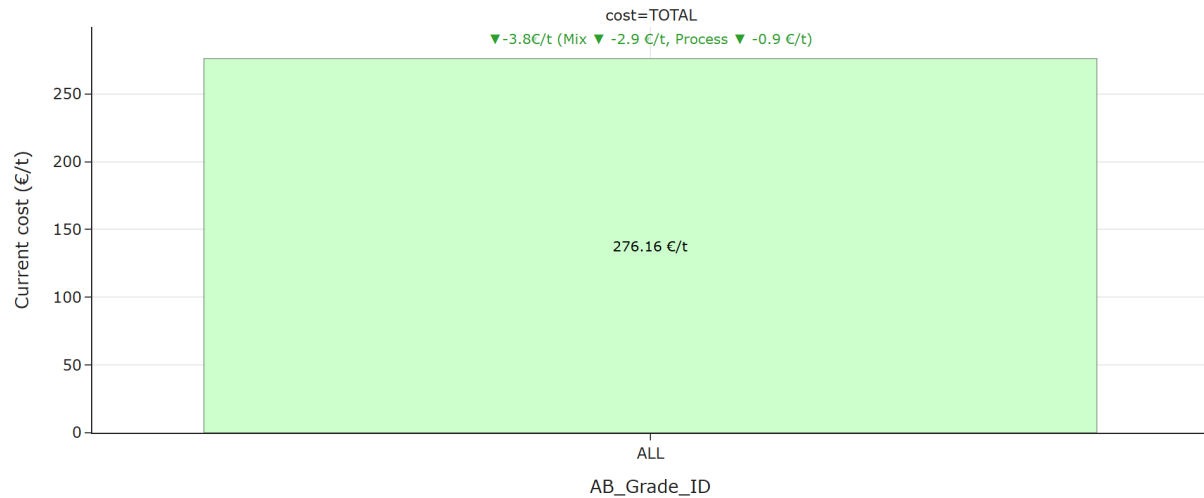
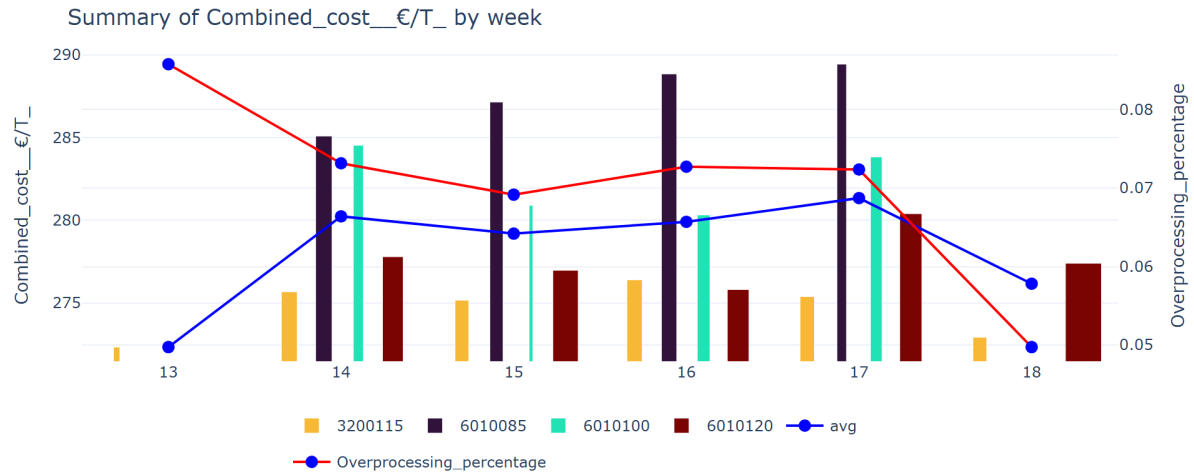
- **Period Analyzed:** March 29, 2026 – May 04, 2026
- **Cost Components Included:** Fibre, steam, electricity, starch, and chemicals.
- **Grades Included:** Only grades 6010085, 6010100, 601020 and 3200115 are included in this week’s analysis.
- **Data Selection Criteria:** Only operating conditions where strength properties exceed specification thresholds were considered.
- **Naming Conventions Used:**
 - **Current:** Refers to the last week of the analyzed period.

- **Historic:** Represents the average cost over the monthly interval previous to the “Current” period. Used as baseline.
- **Best:** Corresponds to the cost associated with the optimal batch configuration.

2 Executive Summary

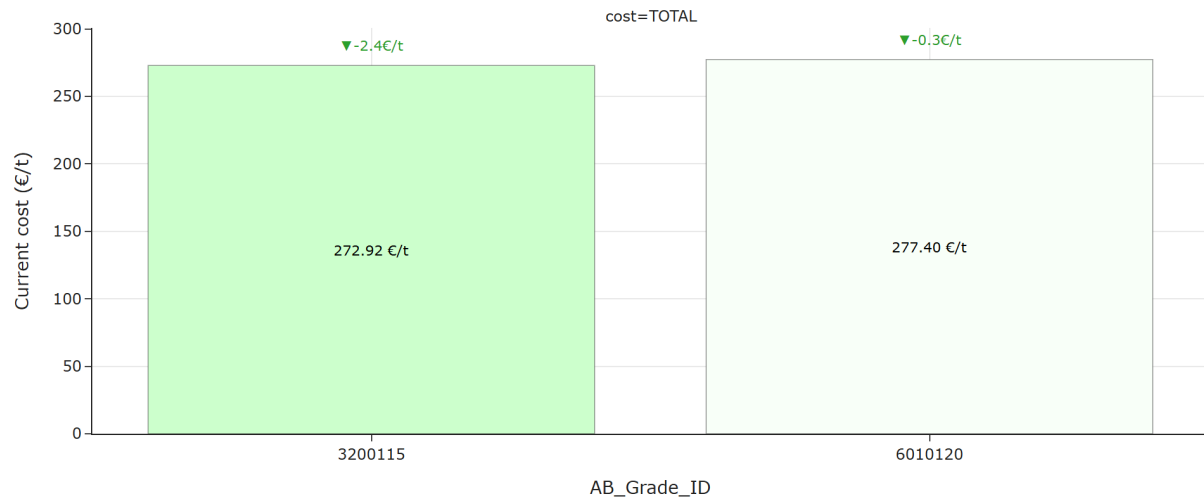
The following plot illustrates how total cost varies across different grades, enabling a comparison between the current configuration and the historical baseline.

The next plot displays how total cost evolves over time, with a breakdown by grade to show individual contributions.



For all considered grades, TOTAL cost saw a decrease of 3.81 €/t, resulting in 276.16 €/t.

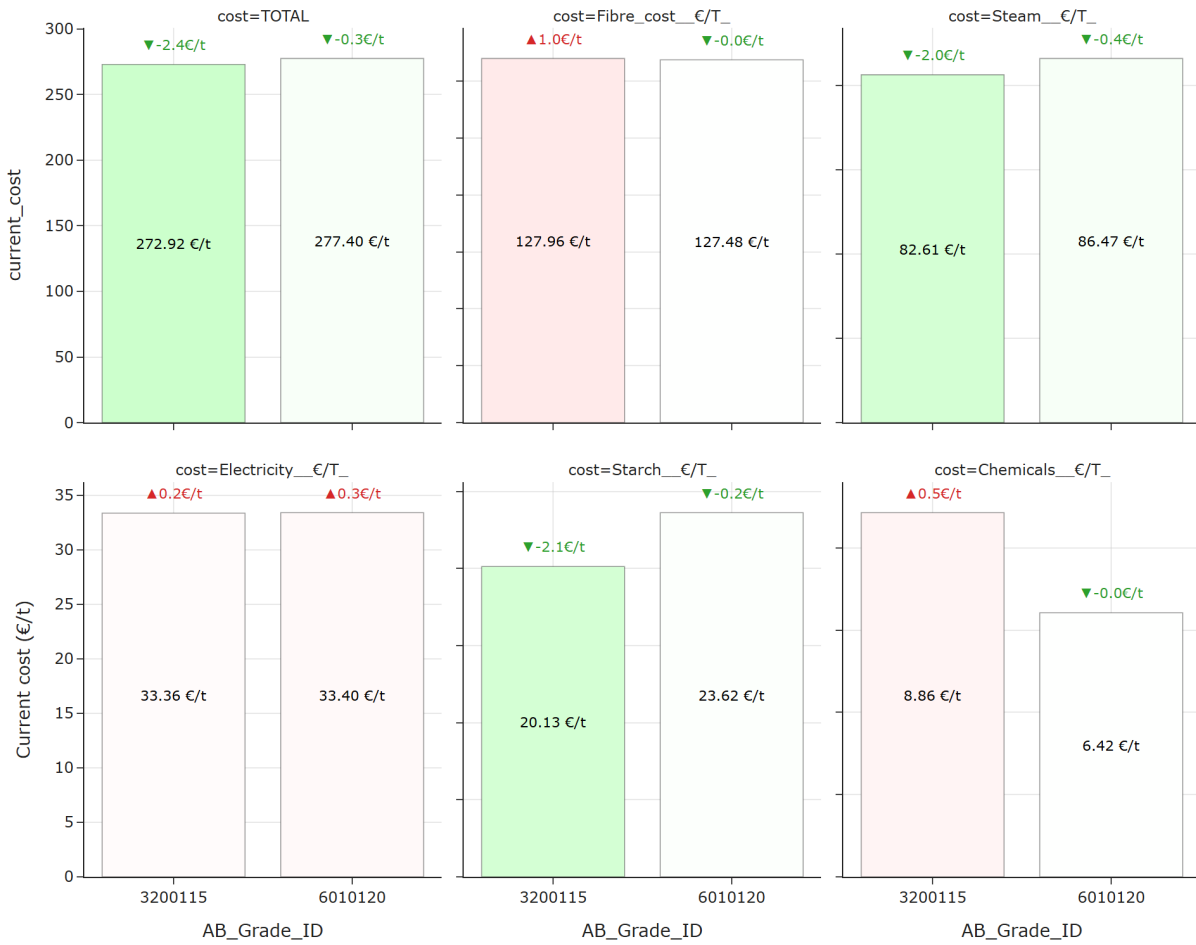
Breaking down the overall cost change, -2.90 €/t was driven by a cheaper mix of grades, and -0.91 €/t was due to an improvement in the process.



For grade 3200115, TOTAL cost saw a decrease of 2.43 €/t, resulting in 272.92 €/t.

For grade 6010120, TOTAL cost saw a decrease of 0.33 €/t, resulting in 277.40 €/t.

Looking across individual grades, the strongest cost decrease was observed for grade 3200115 (-2.43 €/t). These movements stand out and may require further investigation.



Average change by component:

- **Steam_€/T** decreased on average (-1.21 €/t).
- **Starch_€/T** decreased on average (-1.11 €/t).
- **Fibre_cost_€/T** increased on average (+0.48 €/t).
- **Chemicals_€/T** increased on average (+0.24 €/t).
- **Electricity_€/T** increased on average (+0.22 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Fibre_cost_€/T** (+0.48 €/t on average), with the highest increase in grade 3200115 (+0.99 €/t) and the largest average decrease comes from **Steam_€/T** (-1.21 €/t on average), with the biggest drop in grade 3200115 (-2.05 €/t). These components stand out and may require further investigation.



Average change by steam:

- **Steam_Afterdryers** decreased on average (-1.53 €/t).
- **Steam_Predryers** increased on average (+0.54 €/t).
- **Steam_Whitewater** decreased on average (-0.20 €/t).
- **Steam_Starchkitchen** decreased on average (-0.08 €/t).
- **Steam_Heatexchangers** increased on average (+0.05 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Steam_Predryers** (+0.54 €/t on average), with the highest increase in grade 3200115 (+0.59 €/t) and the largest average decrease comes from **Steam_Afterdryers** (-1.53 €/t on average), with the biggest drop in grade 3200115 (-2.93 €/t). These components stand out and may require further investigation.

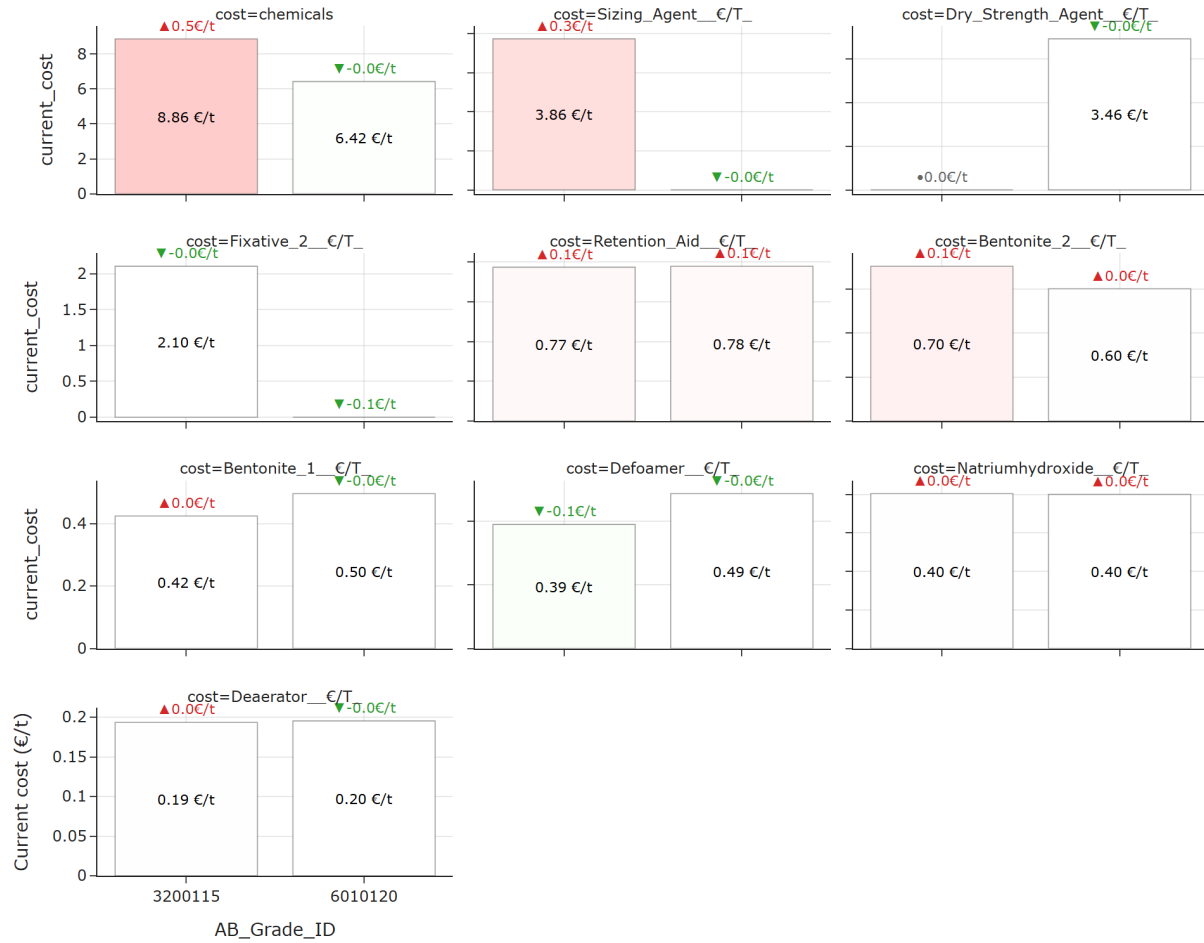


Average change by electricity:

- **Electricity_Specific** increased on average (+0.21 €/t).
- **Electricity_Other** increased on average (+0.18 €/t).
- **Electricity_Afterdryer** decreased on average (-0.08 €/t).
- **Electricity_Wiresection** decreased on average (-0.07 €/t).
- **Electricity_Predryer** decreased on average (-0.05 €/t).
- **Electricity_Freqconverters** increased on average (+0.01 €/t).
- **Electricity_Press_and_Steambox** increased on average (+0.01 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Electricity_Specific** (+0.21 €/t on average), with the highest increase in grade 3200115 (+0.37 €/t) and the largest average decrease comes from **Electricity_Afterdryer** (-0.08 €/t on average),

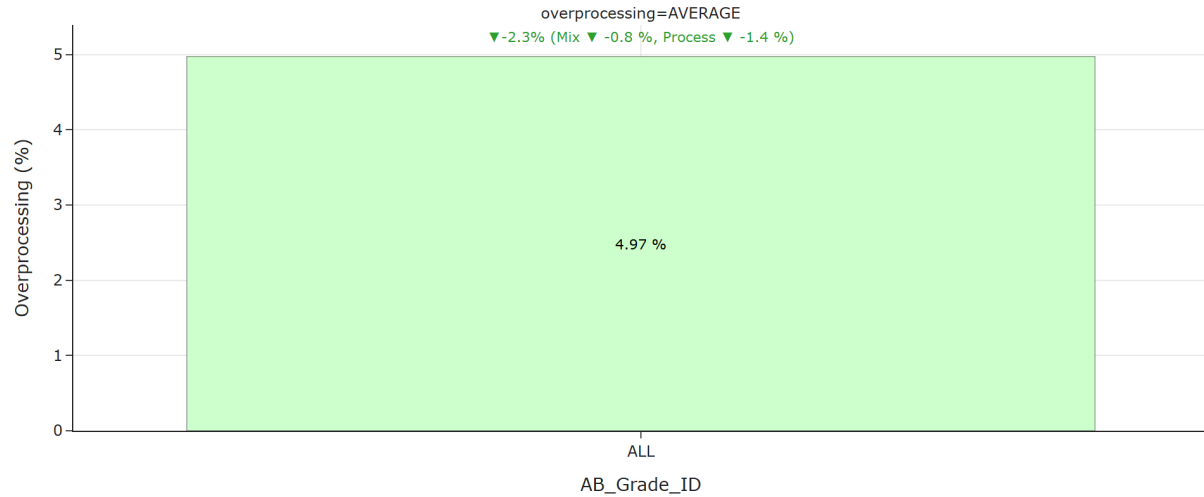
with the biggest drop in grade 3200115 (-0.12 €/t). These components stand out and may require further investigation.



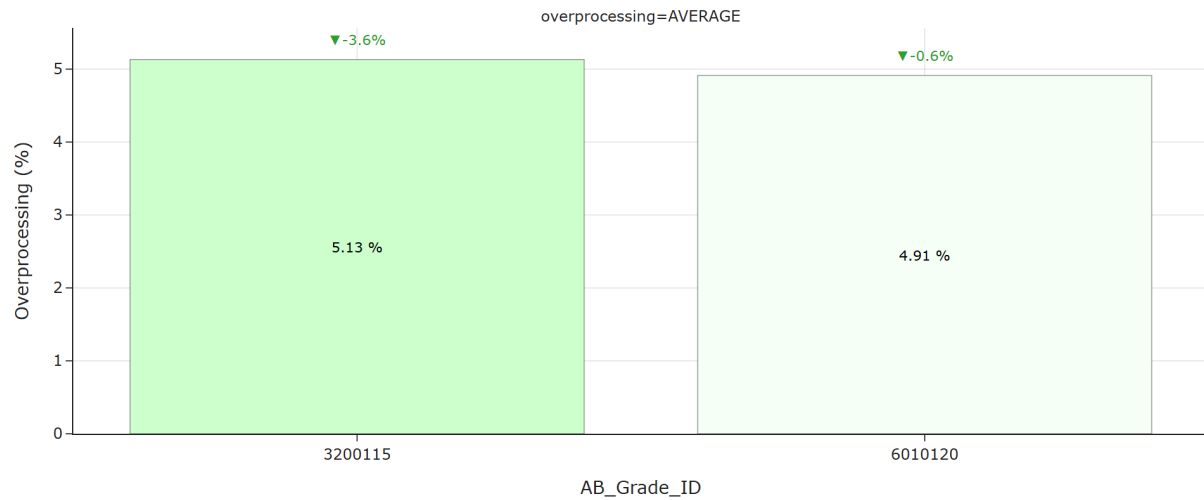
Average change by chemicals:

- **Sizing_Agent_€/T** increased on average (+0.33 €/t).
- **Bentonite_2_€/T** increased on average (+0.08 €/t).
- **Retention_Aid_€/T** increased on average (+0.07 €/t).
- **Defoamer_€/T** decreased on average (-0.03 €/t).
- **Deaerator_€/T** increased on average (+0.01 €/t).
- **Natriumhydroxide_€/T** increased on average (+0.00 €/t).
- **Fixative_2_€/T** decreased on average (-0.00 €/t).
- **Bentonite_1_€/T** increased on average (+0.00 €/t).
- **Dry_Strength_Agent_€/T** decreased on average (-0.00 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Sizing_Agent_€/T** (+0.33 €/t on average), with the highest increase in grade 3200115 (+0.33 €/t) and the largest average decrease comes from **Defoamer_€/T** (-0.03 €/t on average), with the biggest drop in grade 3200115 (-0.06 €/t). These components stand out and may require further investigation.



For all considered grades, TOTAL overprocessing saw a decrease of 2.26 %, resulting in 4.97 %.

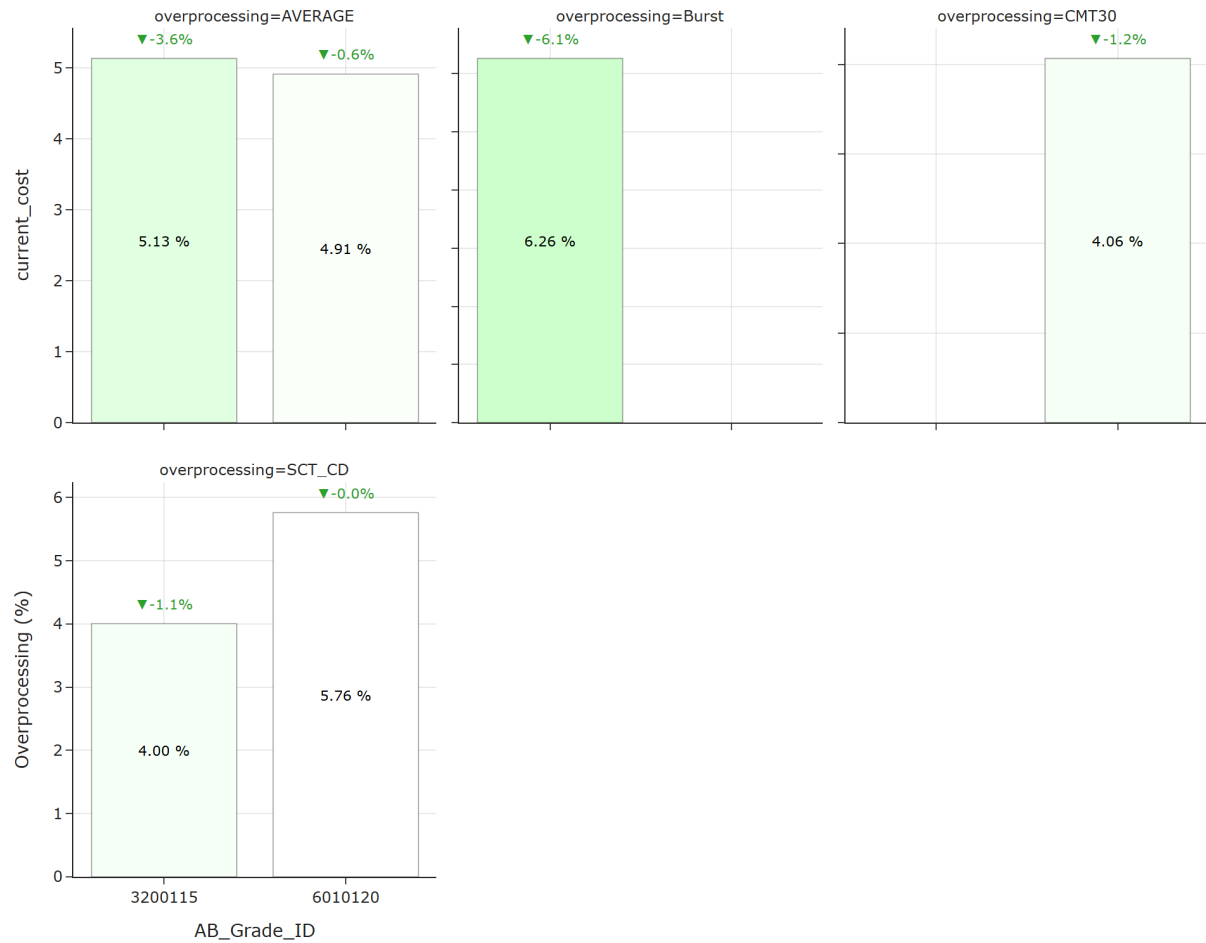


For grade 3200115, TOTAL overprocessing saw a decrease of 3.55 %, resulting in 5.13 %.

For grade 6010120, TOTAL overprocessing saw a decrease of 0.63 %, resulting in 4.91 %.

Looking across individual grades, the strongest overprocessing decrease was observed for grade 3200115 (-3.55 %). These movements stand out and may require further investigation.

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For grade 3200115, TOTAL overprocessing saw a decrease of 3.55 %, resulting in 5.13 %.

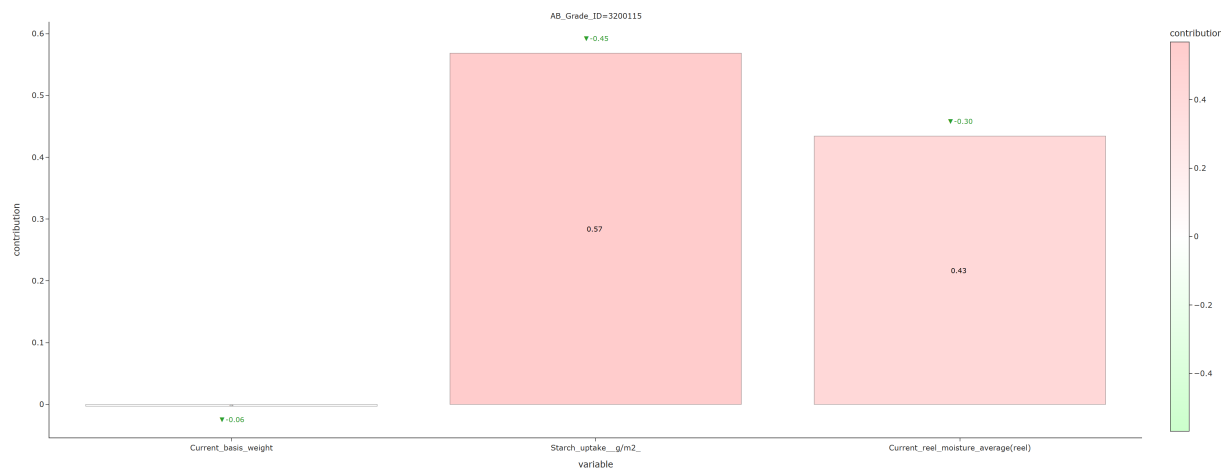
For grade 6010120, TOTAL overprocessing saw a decrease of 0.63 %, resulting in 4.91 %.

Looking across individual grades, the strongest overprocessing decrease was observed for grade 3200115 (-3.55 %). These movements stand out and may require further investigation.

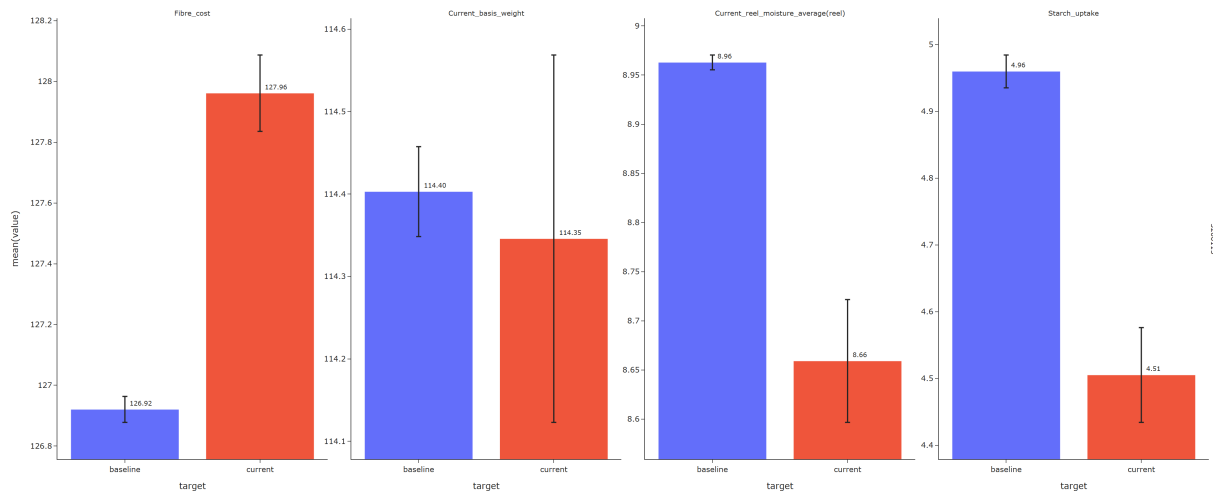
3 Breakdown by grades

3.1 3200115

3.1.1 Fiber cost



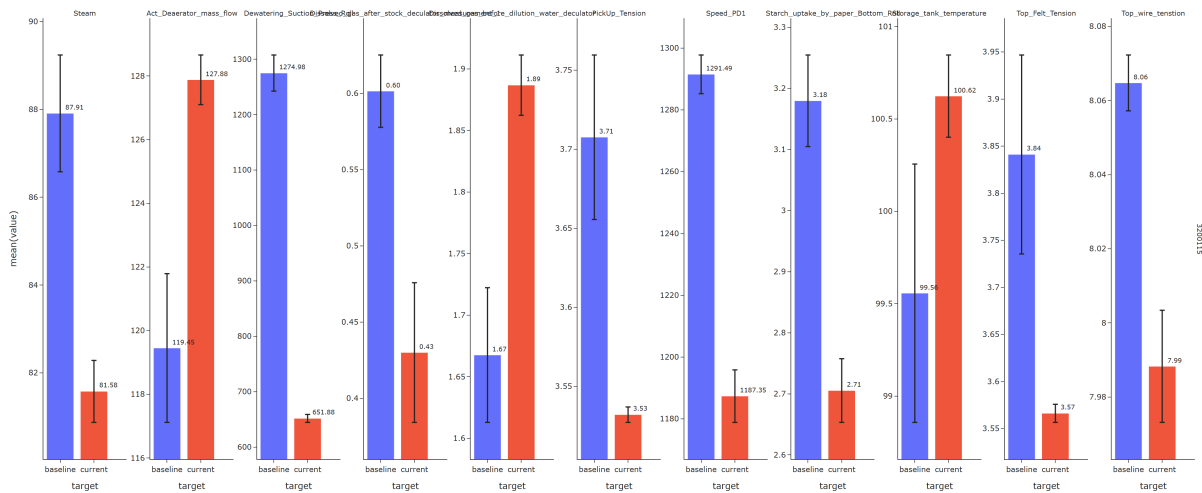
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-increasing drivers: - Starch uptake g/m2 decreased (-0.45)



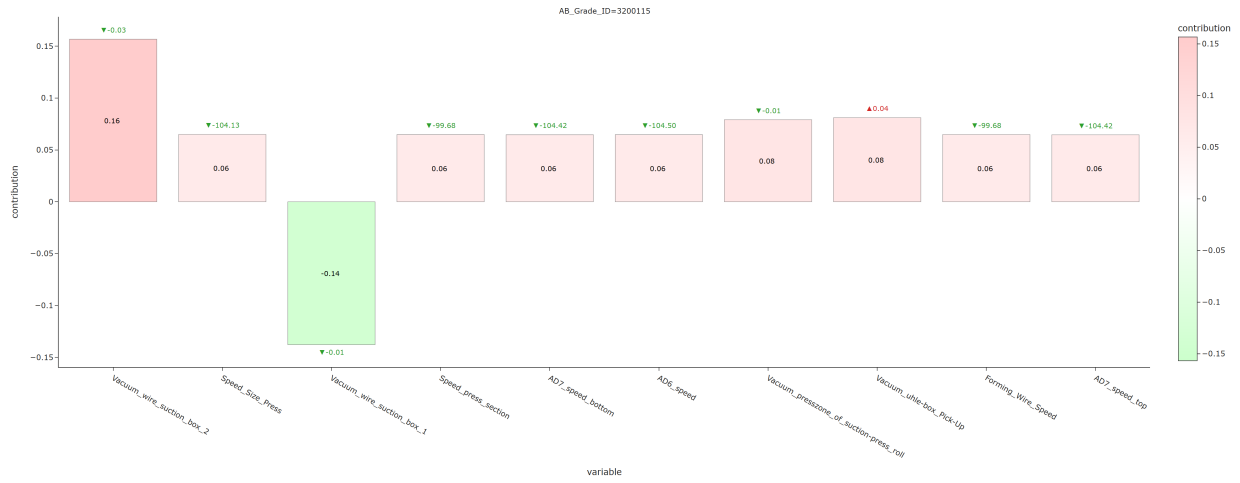
3.1.2 Steam cost



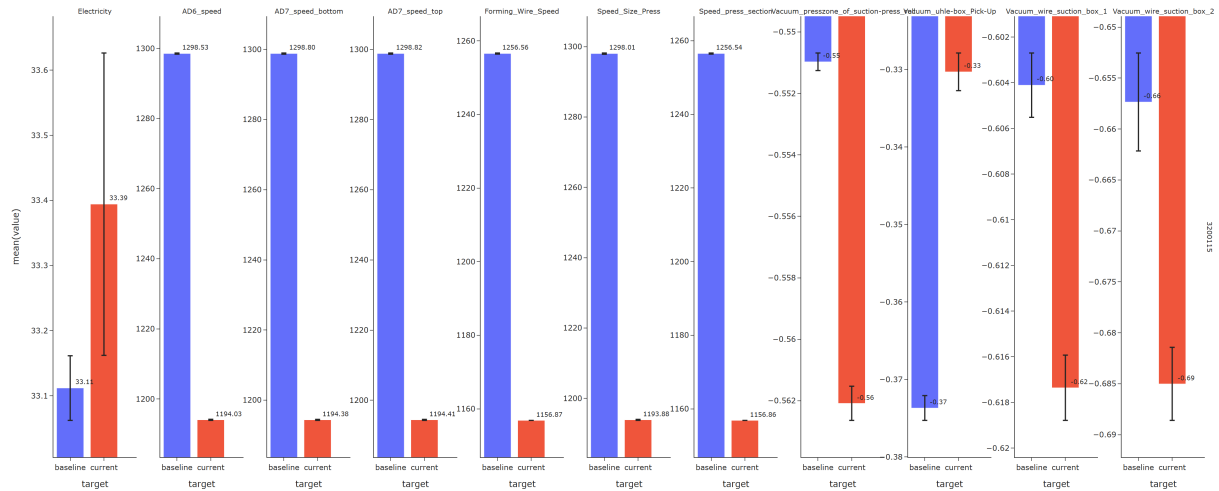
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Storage tank temperature increased (+1.07) - Dissolved gas before dilution water decuator increased (+0.22) Cost-increasing drivers: - Speed PD1 decreased (-104.14) - Dissolved gas after stock decuator measurement 1 decreased (-0.17) - PickUp Tension decreased (-0.18) - Top wire tension decreased (-0.08) - Dewatering Suction Press Roll decreased (-623.10)



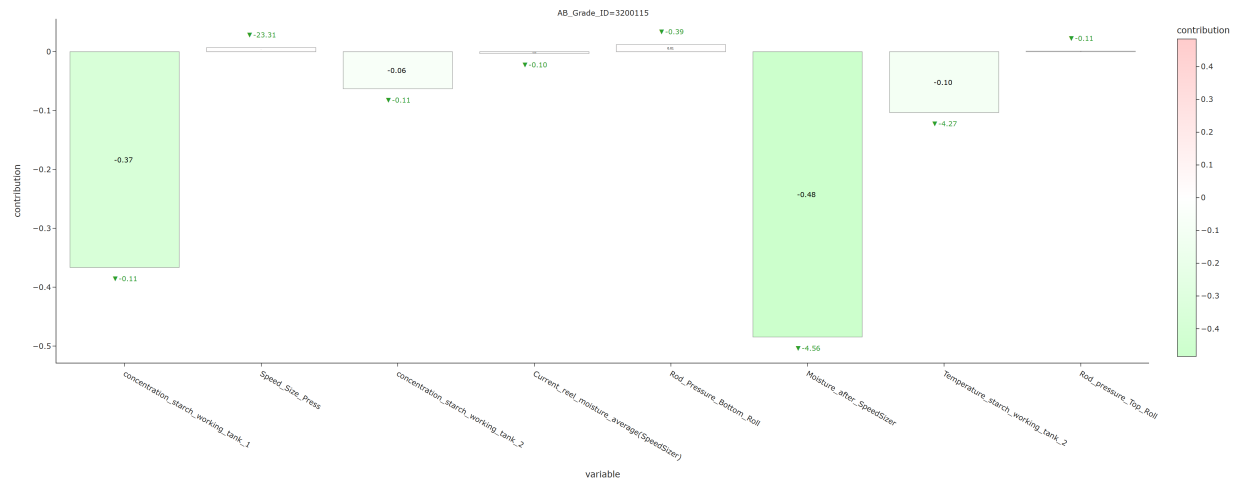
3.1.3 Electricity cost



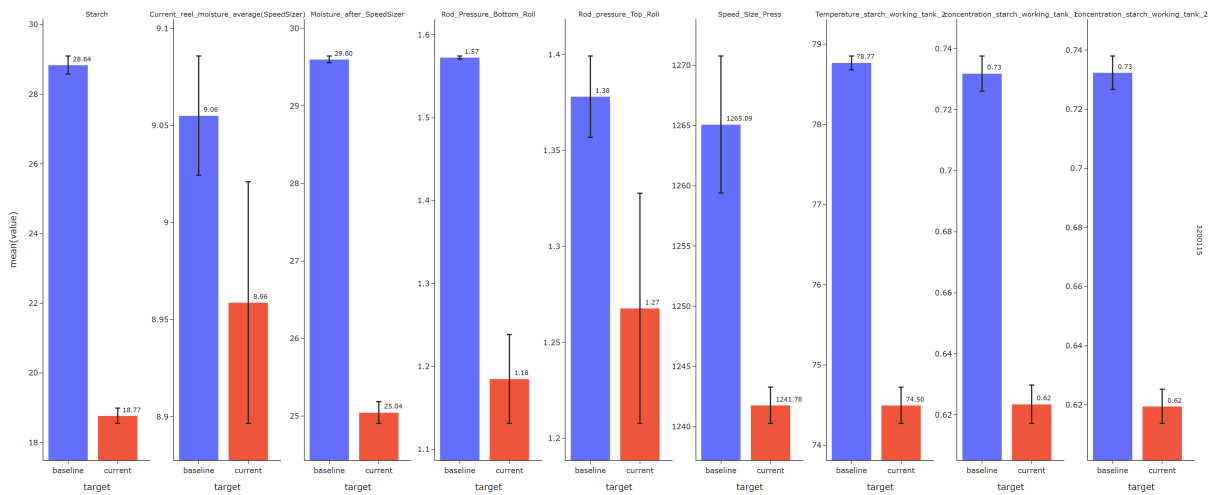
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Vacuum wire suction box 1 decreased (-0.01) Cost-increasing drivers: - Vacuum wire suction box 2 decreased (-0.03) - Vacuum uhle-box Pick-Up increased (+0.04) - Vacuum presszone of suction-press roll decreased (-0.01) - Speed Size Press decreased (-104.13)



3.1.4 Starch cost

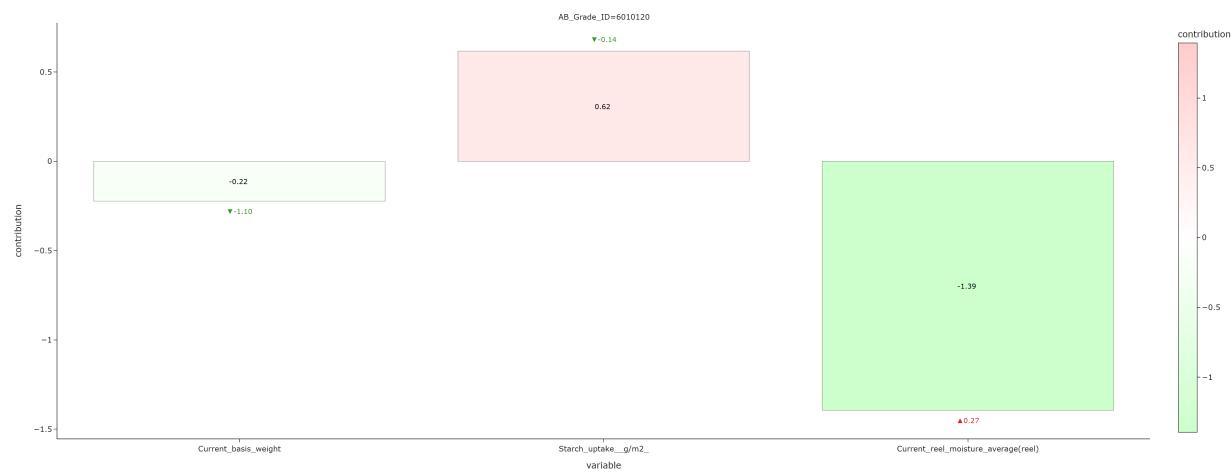


For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Moisture after SpeedSizer decreased (-4.56) - Concentration starch working tank 1 decreased (-0.11)

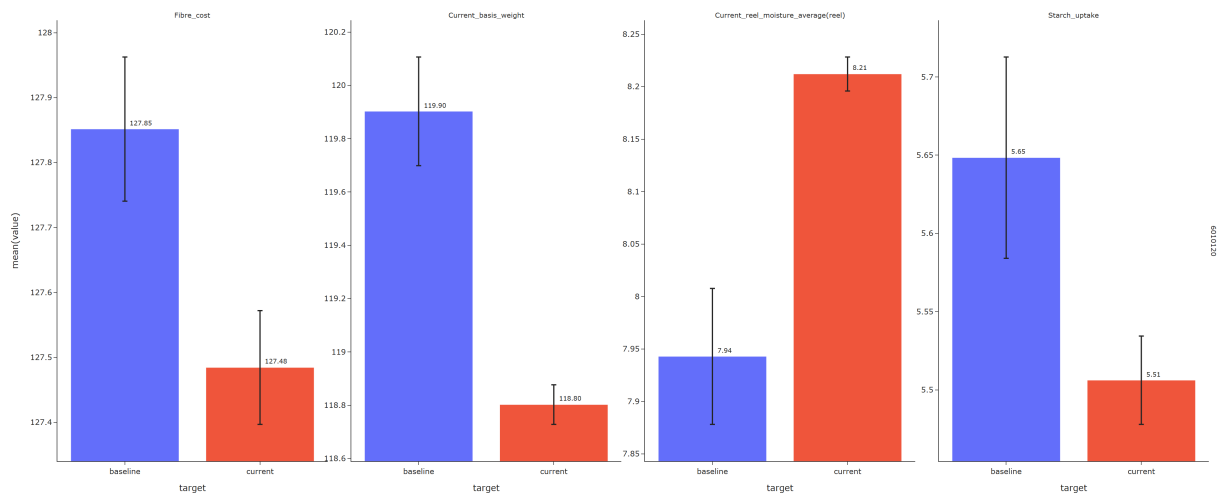


3.2 6010120

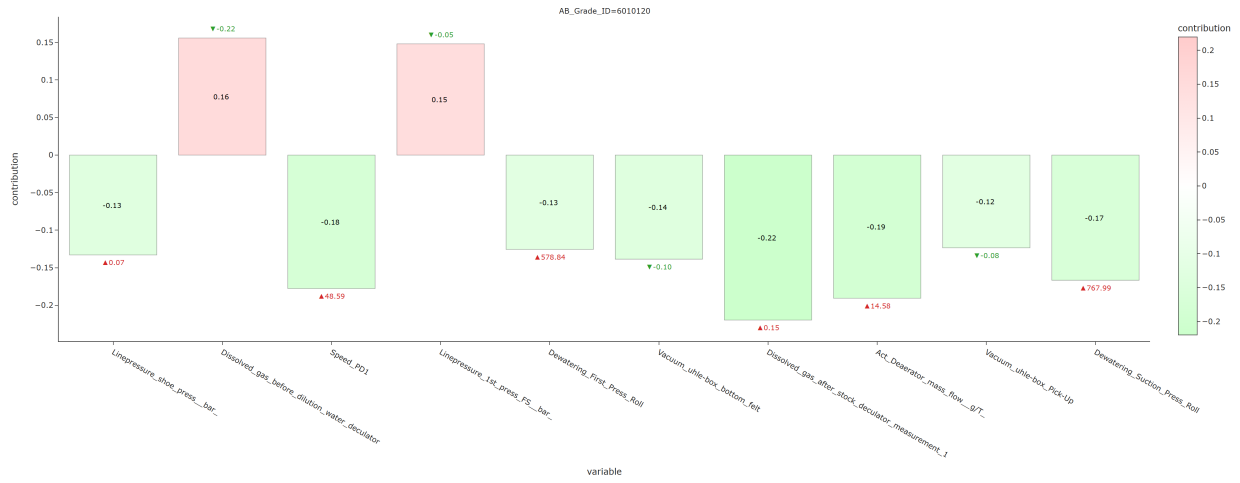
3.2.1 Fiber cost



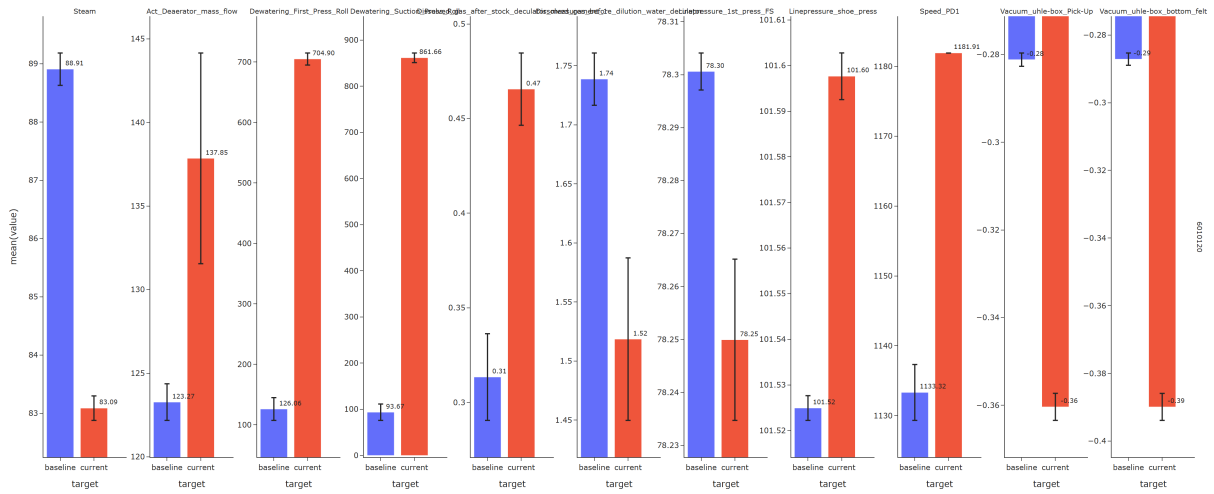
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Current reel moisture average(reel) increased (+0.27)



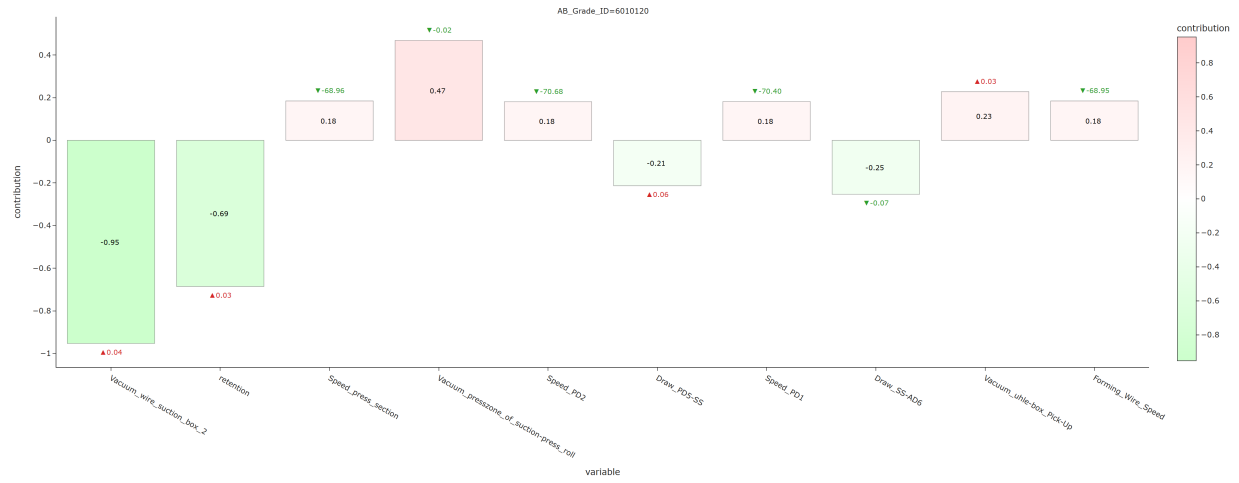
3.2.2 Steam cost



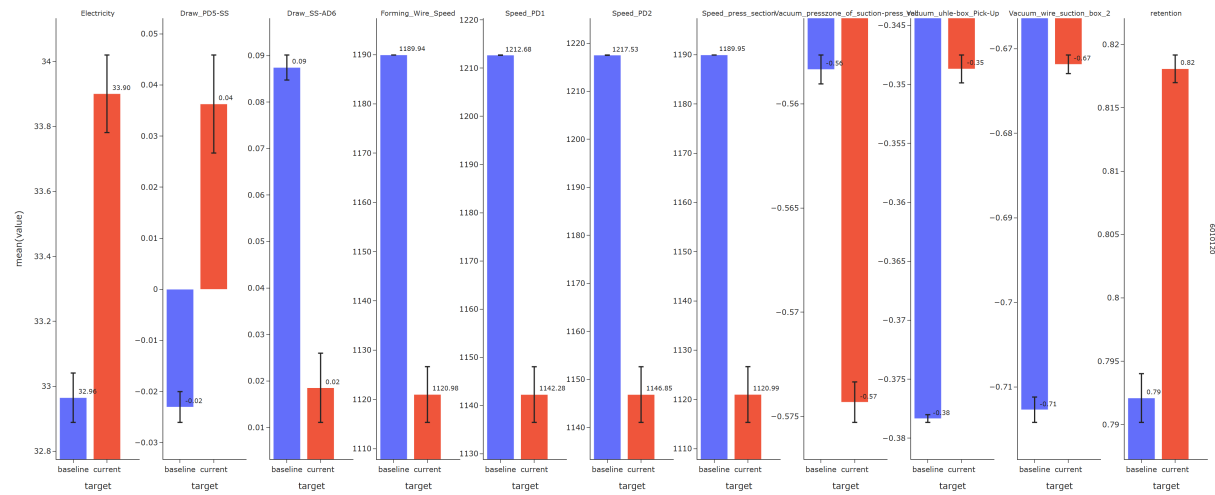
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Dissolved gas after stock deculator measurement 1 increased (+0.15) - Act Deaeator mass flow g/T increased (+14.58) - Speed PD1 increased (+48.59) - Dewatering Suction Press Roll increased (+767.99) Cost-increasing drivers: - Dissolved gas before dilution water decuator decreased (-0.22) - Linepressure 1st press FS bar decreased (-0.05)



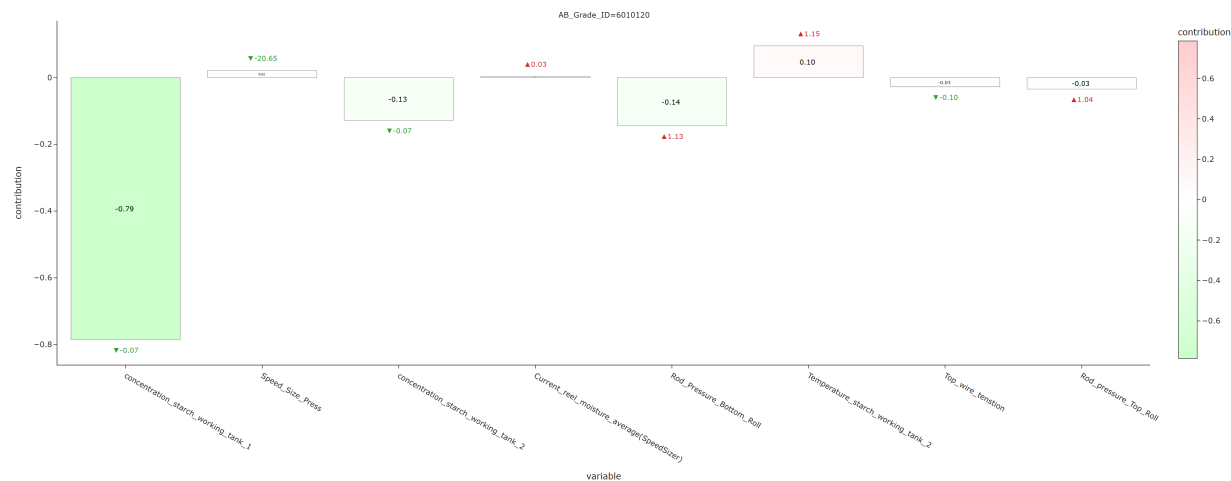
3.2.3 Electricity cost



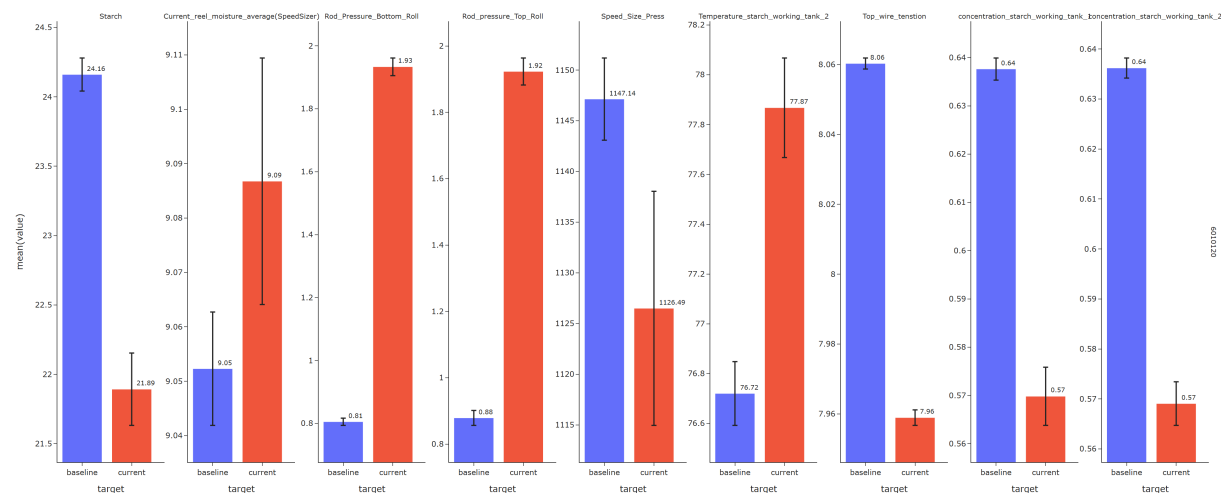
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Vacuum wire suction box 2 increased (+0.04) - Retention increased (+0.03) - Draw SS-AD6 decreased (-0.07) Cost-increasing drivers: - Vacuum presszone of suction-press roll decreased (-0.02) - Vacuum uhle-box Pick-Up increased (+0.03)



3.2.4 Starch cost



For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Concentration starch working tank 1 decreased (-0.07) - Rod Pressure Bottom Roll increased (+1.13)



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4 Summary, Diagnosis, and Action Plan

4.1 Executive Summary

The mill achieved a **3.81 €/t reduction in total production costs** (to 276.16 €/t), driven by both improved grade mix (-2.90 €/t) and process optimization (-0.91 €/t). Overall overprocessing decreased by 2.26% to 4.97%. However, the analysis reveals significant variability between grades and competing cost drivers that require targeted intervention.

Grade 3200115 shows the most dramatic improvements but also concerning process instabilities, while **Grade 6010120** demonstrates more modest but stable gains.

4.2 Detailed Diagnosis

4.2.1 Grade Performance Analysis

4.2.1.1 Grade 3200115 (Strongest Performer) **Cost Section:** Total cost decreased 2.43 €/t to 272.92 €/t **Quality Section:** Overprocessing reduced 3.55% to 5.13%

Critical Issues Identified: - **Drying Section:** Major steam reduction in afterdryers (-2.93 €/t) offset by predryer increases (+0.59 €/t), suggesting heat distribution imbalance [papermaking_science_technology.pdf:d6f6f20ccb65b658:4629:0 describes drying section steam consumption patterns] - **Speed Instability:** PD1 speed decreased 104.14 units and Size Press speed decreased 104.13 units - this synchronized reduction indicates a grade change or production disruption [papermaking_science_technology.pdf:d6f6f20ccb65b658:4601:0 discusses grade change impacts on machine speed] - **Press Section:** Dewatering suction press roll decreased 623.10 units, severely impacting mechanical dewatering efficiency

4.2.1.2 Grade 6010120 (Stable Performer) **Cost Section:** Total cost decreased 0.33 €/t to 277.40 €/t **Quality Section:** Overprocessing reduced 0.63% to 4.91%

Positive Indicators: - Speed PD1 increased 48.59 units - Dewatering suction press roll increased 767.99 units - Improved moisture control (reel moisture increased 0.27%)

4.3 Component-Level Analysis

4.3.1 Raw Materials Section

Fiber Cost (Furnish): +0.48 €/t average increase - Grade 3200115: +0.99 €/t (concerning) - Likely related to furnish composition changes during grade transitions [papermaking_science_technology.pdf:d6f6f20ccb65b658:4602:0 notes furnish changes affect dewatering and overall process]

Starch (Surface Treatment): -1.11 €/t average decrease - Grade 3200115: Starch uptake decreased 0.45 g/m² (cost-increasing driver) - Concentration in working tank 1 decreased 0.11% (cost-reducing) - Indicates improved starch kitchen efficiency but potential quality concerns

4.3.2 Energy Section - Steam (Drying)

Afterdryers: -1.53 €/t average (largest decrease) - Grade 3200115: -2.93 €/t reduction - **Diagnosis:** Improved moisture profile entering afterdryer section; moisture after SpeedSizer decreased 4.56% for Grade 3200115

Predryers: +0.54 €/t average (largest increase) - Grade 3200115: +0.59 €/t - **Diagnosis:** Compensating for reduced mechanical dewatering in press section [papermaking_science_technology.pdf:d6f6f20ccb65b658:4629:0 shows press section impacts drying energy]

Other Steam Components: - Whitewater: -0.20 €/t (stock preparation efficiency) - Starch kitchen: -0.08 €/t (process optimization) - Heat exchangers: +0.05 €/t (minimal impact)

4.3.3 Energy Section - Electricity

Specific Electricity: +0.21 €/t average increase - Grade 3200115: +0.37 €/t - **Diagnosis:** Increased refining or pumping loads

Afterdryer Electricity: -0.08 €/t decrease - Correlates with reduced steam consumption

Other Components: Mixed minor changes in wire section, press, predryer (all <0.10 €/t)

4.3.4 Chemicals Section (Wet End/Additives)

Sizing Agent: +0.33 €/t (largest chemical increase) - Grade 3200115: +0.33 €/t - May compensate for reduced starch uptake

Retention System: - Retention Aid: +0.07 €/t - Bentonite 2: +0.08 €/t - Grade 6010120 showed retention increase of 0.03% (cost-reducing) - [papermaking_science_technology.pdf:d6f6f20ccb65b658:4602:0 notes retention changes during grade transitions]

Defoamer: -0.03 €/t (improved process stability)

4.4 Root Cause Analysis

4.4.1 Grade 3200115 Critical Factors

Wet End/Forming Section: - Dissolved gas management issues: decreased after stock deculator (-0.17), increased before dilution water deculator (+0.22) - Vacuum system imbalances: wire suction box 2 decreased, uhle-box pickup increased - **Impact:** Poor sheet formation and dewatering

Press Section: - Massive dewatering suction press roll decrease (-623.10) - Pickup tension decreased (-0.18) - Top wire tension decreased (-0.08) - Line pressure 1st press decreased - **Impact:** Inadequate mechanical dewatering, forcing thermal drying compensation

Size Press Section: - Speed decrease (-104.13) synchronized with PD1 - Moisture after SpeedSizer decreased (-4.56%) - POSITIVE - **Impact:** Better moisture management but production loss

Stock Preparation: - Storage tank temperature increased (+1.07°C) - POSITIVE for chemical reactions - Deaerator mass flow changes affecting dissolved gas levels

4.4.2 Grade 6010120 Optimization Drivers

- Improved dewatering (suction press roll +767.99)
 - Better speed management (PD1 +48.59)
 - Enhanced retention (+0.03%)
 - Optimized moisture profile (reel moisture +0.27%)
-

4.5 Action Plan

4.5.1 Immediate Actions (0-2 weeks)

4.5.1.1 Grade 3200115 - Press Section Emergency

1. Investigate press section dewatering failure

- Check suction press roll vacuum levels and felt condition
- Verify press loading and nip pressure settings
- Inspect felt conditioning and shower systems
- **Target:** Restore dewatering capacity to reduce predryer steam by 0.50 €/t

2. Speed stability investigation

- Analyze PD1 and Size Press speed correlation
- Review grade change procedures [papermaking_science_technology.pdf:d6f6f20ccb65b658:4601:0]
- Identify if this is grade change artifact or operational issue
- **Target:** Eliminate unnecessary speed variations

3. Dissolved gas management

- Calibrate dissolved gas analyzers
- Check deculator and deaerator operation
- Optimize deaerator dosing (currently +14.58 g/T for Grade 6010120)
- **Target:** Stabilize dissolved gas at <8 ppm

4.5.1.2 Both Grades - Fiber Cost Control

4. Furnish optimization review

- Analyze fiber cost increase drivers (+0.99 €/t for Grade 3200115)
- Review virgin/recycled fiber ratios
- Evaluate broke management system
- **Target:** Reduce fiber cost by 0.30 €/t through mix optimization

4.5.2 Short-term Actions (2-8 weeks)

4.5.2.1 Drying Section Optimization

5. Steam distribution rebalancing

- Implement dynamic steam pressure control for grade changes
- Optimize predryer/afterdryer heat distribution
- **Target:** Reduce total steam cost by additional 0.40 €/t
- **Reference:** [papermaking_science_technology.pdf:d6f6f20ccb65b658:4629:0] for benchmarking

6. Moisture profile optimization

- Leverage Grade 3200115 success (moisture after SpeedSizer -4.56%)
- Apply learnings to Grade 6010120
- Install/calibrate additional moisture sensors if needed
- **Target:** Reduce overprocessing below 4.5% for both grades

4.5.2.2 Chemical System Optimization

7. Sizing agent efficiency study

- Investigate +0.33 €/t increase
- Correlate with starch uptake reduction (-0.45 g/m²)
- Test alternative sizing strategies
- **Target:** Reduce sizing agent cost by 0.15 €/t while maintaining quality

8. Retention system tuning

- Optimize retention aid and bentonite dosing
- Implement retention monitoring and control
- **Target:** Improve first-pass retention by 1%, saving 0.10 €/t in chemicals and fiber

4.5.3 Medium-term Actions (2-6 months)

4.5.3.1 Process Control Enhancement

9. Automated grade change system

- Implement closed-loop grade change control [papermaking_science_technology.pdf:d6f6f20ccb65b658:4629:0]
- Develop dynamic models for steam pressure adjustment
- Create grade-specific parameter sets
- **Target:** Reduce grade change time by 20%, minimize transient losses

10. Vacuum system upgrade (Grade 3200115)

- Balance wire section vacuum distribution
- Optimize uhle-box and suction box settings
- Consider felt renewal program
- **Target:** Improve sheet dryness after press by 2%, saving 0.60 €/t in steam

4.5.3.2 Energy Management

11. Electricity consumption analysis

- Investigate specific electricity increase (+0.21 €/t average)
- Benchmark against industry standards [papermaking_science_technology.pdf:d6f6f20ccb65b658:4629:0]
- Identify motor and drive inefficiencies
- **Target:** Reduce specific electricity by 0.15 €/t

12. Heat recovery optimization

- Evaluate heat exchanger performance (+0.05 €/t increase)
- Assess whitewater heat recovery potential
- **Target:** Improve heat recovery efficiency by 5%

4.5.4 Long-term Strategic Actions (6-12 months)

4.5.4.1 Grade Strategy Optimization

13. Grade mix analysis

- Current mix savings: -2.90 €/t
- Analyze profitability by grade considering all cost factors
- Develop optimal production scheduling
- **Target:** Additional 0.50 €/t savings through strategic grade mix

14. Quality vs. Cost optimization

- Current overprocessing: 4.97% (down from 7.23%)
- Implement statistical process control [papermaking_science_technology.pdf:d6f6f20ccb65b658:6951:0]
- Balance quality specifications with production costs
- **Target:** Reduce overprocessing to 3.5%, saving 0.80 €/t

4.5.4.2 Capital Investment Evaluation

15. Press section upgrade feasibility

- Evaluate ROI for press section modernization
- Consider shoe press or extended nip technology
- **Potential:** 2-3% dryness improvement = 1.50-2.00 €/t savings
- **Reference:** [papermaking_science_technology.pdf:d6f6f20ccb65b658:3395:0] for cost-benefit methodology

16. Advanced process control (APC)

- Implement model predictive control for drying section
- Deploy real-time optimization for chemical dosing
- **Potential:** 0.50-1.00 €/t additional savings

4.6 Key Performance Indicators (KPIs) to Monitor

4.6.1 Cost Metrics

- Total cost per ton (target: <270 €/t)
- Steam cost per ton (target: reduce by additional 0.50 €/t)
- Fiber cost per ton (target: <current - 0.30 €/t)
- Chemical cost per ton (target: stabilize at current)